When Correctness Is the Prediction: An Evaluation Trap in Sampling-Based Uncertainty for Handwritten Mathematics

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Abstract

Sampling a vision–language model repeatedly and measuring its self-disagreement yields a cheap, label-free uncertainty signal. We evaluate this signal on handwritten mathematics and report a positive result, a negative result, and a mechanism by which the evaluation itself manufactures an effect that is not present. On transcription the signal is effective: AUROC 0.835*, replicated at* 0.828 *on a second, architecturally independent model family, robust in both cases to every artifact control we apply, and superior by resolved margins to both the model's own token confidence and to simpler reductions of the same samples. A rule that abstains whenever all five samples disagree identifies a genuinely incorrect transcription* 57 *times out of* 61*. On grading the signal is absent: AUROC* 0.520 *on a balanced sample, statistically indistinguishable from chance. The methodological risk lies between these two results. Stratifying the grading task by its ground-truth label, a natural response to a model exhibiting strong answer bias, appears to recover a substantial effect (*0.801–0.854*) that replicates across three independently trained model families. That effect is an artifact. Within a single-label stratum, "was the model correct" is the same column as "what did the model answer", so an uncertainty measure computed from those answers predicts correctness almost by construction; a signal-free biased coin attains a* higher *AUROC than any model evaluated here. We report the result we initially accepted, the diagnostic that retired it, and the property of the transcription task under which the same measure remains sound. Finally, a complete human audit of the scorer's own verdicts introduces a third caution: approximately three-quarters of the items it marks incorrect are attributable to the scoring pipeline rather than to the model, so the correctness label against which these AUROCs are measured is itself noisy.*

1. Introduction

Automated marking of handwritten mathematics fails silently. When a model misreads 4.9 as 49, the resulting transcription is as fluent, as confident and as well-formed as a correct one, and nothing in the output distinguishes the two. In a deployment where that output contributes to a grade, determining *when* to route a page to a human reviewer is more valuable than a marginal gain in average accuracy, and unlike accuracy it is directly actionable.

Repeated sampling, combined with a measure of the resulting self-disagreement, is the natural candidate for such a signal. It requires no training, no labels and no access to model weights, and therefore applies to closed models served behind an API [6, 13]. Whether it is effective remains an empirical question. We address that question on FERMAT [9], a benchmark of authentic student solutions rendered as handwritten pages, each carrying a deliberately injected error.

Effectiveness proves to be task-dependent, and, more importantly for evaluation practice, the headline figure is easy to misestimate in either direction. On transcription the signal is genuine and survives every control we apply. On grading it is absent. Between these two findings lies a methodological trap. Confronted with a model whose grading verdict is heavily biased toward a single answer, the natural response is to stratify by the ground-truth label. Doing so appears to recover a substantial effect that replicates across three independently trained model families. That effect is an artifact: within a stratum in which the true label is constant, "was the model correct" is arithmetically the same column as "what did the model answer", so an uncertainty measure computed from those answers predicts correctness almost by construction. Consequently, a signal-free biased coin outscores every model evaluated here. Notably, none of the usual safeguards detected the problem. The pre-registered threshold did not, since the null distribution clears it; statistical power did not, since both strata were adequately powered; and replication did not, since it reproduced the artifact three times.

A second and more easily overlooked failure underlies both arms. The automatic scorer that assigns correctness is itself unreliable. A complete human audit of every item it marked incorrect finds that approximately three-quarters of those verdicts are attributable to the scoring pipeline rather than to the model, while a random audit of the items it marked correct reveals unearned passes in the opposite direction. We quantify the consequences of this label noise, and delimit what it does not affect.

This is an evaluation paper rather than a method paper. We propose no new uncertainty estimator. Instead, we establish where a standard estimator is effective, where it is not, and two distinct mechanisms by which an evaluation of it can yield a confident figure that carries no information.

2. Related Work

Sampling-based uncertainty. Repeated sampling turns a generator into an ensemble, and disagreement among the samples is used as a confidence signal. Self-consistency aggregates sampled chains by majority vote and improves accuracy without any confidence estimate [13]; semantic entropy makes the uncertainty explicit by clustering samples that mean the same thing and taking the entropy over clusters [6]. We inherit the resampling and the entropy, but not the clustering: our clusters are exact-match over an *extracted answer span*, not entailment-based. That choice is not an oversight. On this data an NLI-plus-union-find implementation merged opposed clusters through a single false-positive transitive link and degraded as K grew, so the simpler rule is the one that survives (§6). A separate line asks the model to report its own confidence, either as a probability of being correct [5] or in words; we test the verbalized form as a baseline and it carries no signal on the arm we lead with (§4.4).

Selective prediction and abstention. The decision this signal supports is deferral, not correction, which places the work in selective prediction: a model may abstain, and the question is the risk–coverage trade-off it achieves [4]. We report the same currency, AURC against a random-deferral baseline, and add coverage at a target risk. Abstention has been studied in vision–language models directly [12], the closest prior work to ours in setting: it reduces unnecessary abstention on visual question answering over natural images. The task here differs in a way that matters for the metric: we score dense transcription of a handwritten page against a reference that is itself deliberately perturbed, so "correct" means faithful to a page that may be wrong, not mathematically true.

Vision–language models on handwritten and mathematical content. Visual mathematics benchmarks have grown quickly, but most evaluate *answer accuracy* on typeset figures [8], and document transcription is usually treated as a solved sub-problem with a single reference output [3]. FERMAT [9] is the exception we build on: it renders real student solutions as handwritten pages and injects a controlled error into each, so the reference answer is deliberately wrong in a known way. ErrorRadar [14] and ScratchMath [11] share the error-detection framing; neither contains correct solutions, which (§4.6) makes both single-label strata on which our measurement degenerates. What all three measure is whether a model finds the error. What none measures is whether the model *knows when it has failed to read the page*, which is the question here.

Evaluation artifacts. Our central negative result is that a metric can manufacture a signal from nothing, and this has precedent worth naming plainly. Apparent emergent abilities have been shown to be an artifact of the metric rather than the model [10]. Our case is narrower and, we think, sharper: within a stratum where the true label is constant, correctness becomes the same column as the model's own verdict, so a signal-free biased coin outscores every model we tested (§4.6). The scoring machinery is a second source of artifacts. Step-level verification of mathematical solutions is an active area [7], and automatic judges are increasingly used in place of exact match. Such judges, however, inherit biases of their own [15]. We find a specific failure of that kind: on a perturbed-answer benchmark, judges reconstruct the mathematically correct answer and grade against *that* rather than against the reference they were given (§6).

3. Setup

Data. FERMAT [9] contains $\sim$2,200 photographed handwritten solutions to middle- and high-school problems, in which errors were deliberately introduced by the dataset authors and annotated by type. Each image contains a complete question and its answer, and 15% of items are error-free. That last property is load-bearing for this paper: §4.8 shows the analysis in §4.5 is impossible on datasets that lack it. Unless stated otherwise we use a balanced $n = 300$ sample (150 with an error, 150 without), drawn once with a fixed seed.

Tasks. We evaluate two distinct tasks on the same images. *Perception* asks the model to "explicitly perform OCR on the handwritten text and extract the content in LaTeX format"; it is correct when the extracted final answer matches the reference after canonicalisation. *Reasoning* asks it to "analyze the Answer to determine whether there is any error" and emit a binary judgement; it is correct when that judgement matches the annotation. The model is never shown the reference answer.

Uncertainty measure. For each item we draw $K = 5$ samples at temperature 0.7, reduce each to a canonical label, and take the Shannon entropy of the resulting cluster distribution, $H = -\sum_i p_i \ln p_i$. $H = 0$ when all samples agree and $H = \ln K$ when all differ. We deliberately use exact-match clustering over an *extracted answer span* rather than semantic clustering of whole derivations; §6 records why.

Models. Qwen2.5-VL at 3B and 7B [2], Pixtral-12B [1], LLaVA-NeXT-7B and InternVL3-8B, comprising four independently trained families, all run in bfloat16.

Statistics. Every AUROC carries a 95% bootstrap interval over $10{,}000$ item-level resamples. Comparisons between conditions use a *paired* bootstrap over the same resampled items rather than checking whether two marginal intervals overlap. Before any run we registered a minimum minority-class size of 30 and, for the stratified reasoning hypothesis, an AUROC threshold of 0.70; verdicts below are assigned by a pure function of those thresholds, not by inspection.

4. Results

4.1. Does self-disagreement predict transcription error?

Yes, and by a margin that does not depend on the model's own confidence being poor. On the balanced $n = 300$ sample, perception entropy predicts transcription error with AUROC 0.835 $[0.787, 0.879]$. The model's own mean token log-probability reaches only 0.537 $[0.469, 0.605]$, an interval that includes chance, and the paired difference is $+0.297$ $[+0.214, +0.380]$, resolved. The signal is therefore not recoverable from information the model already exposes.

Read as a deferral rule the result is directly usable: of the 61 items on which all five transcriptions disagree, 57 are wrong (93.4% precision, 31.3% recall). Over the full risk–coverage curve this gives AURC 0.410 against a 0.607 random-deferral baseline.

4.2. Does it replicate on a second model family?

Yes. Pixtral-12B, architecturally independent of Qwen, reaches 0.828 $[0.782, 0.871]$ on the same 300 images at a comparable transcription accuracy (41.7% against 39.3%). The intervals overlap along almost their whole length, so the two are not resolvably different.

More important than the point estimate is that it survives the control of §4.4: 0.772 $[0.710, 0.833]$ after removing both parse failures and maximum-entropy items, with 17% of items at the ceiling. §4.9 describes a fourth model that scores *higher* raw and does not survive that control, which is the reason we report the two together.

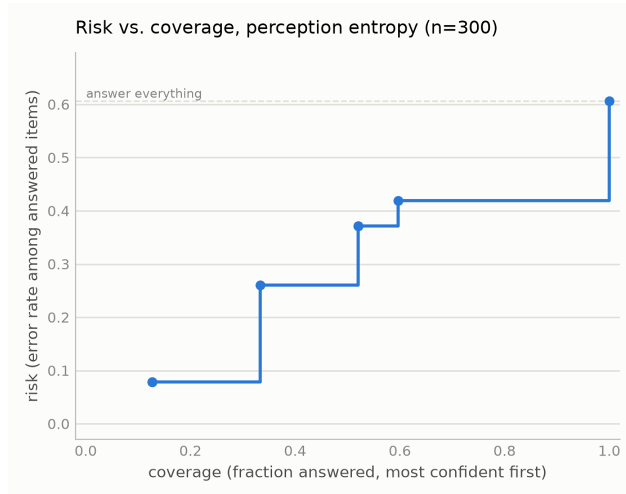

Figure 1. Risk–coverage for perception entropy, Qwen2.5-VL-3B, $n = 300$. Deferring the highest-entropy items first traces AURC 0.410 against a 0.607 random-deferral baseline. The curve is the practitioner-facing form of the result: at a 10% risk target the rule retains 21.3% coverage, and token log-probability cannot hold error below 40% at any coverage.

Signal	AUROC (95% CI)	vs. entropy
Sampling entropy	0.835 $[0.787, 0.879]$	n/a
$1-$majority fraction	0.793 $[0.742, 0.841]$	$+0.042$ $[+0.011, +0.073]$
Distinct count	0.790 $[0.739, 0.838]$	$+0.045$ $[+0.016, +0.075]$
Majority margin	0.779 $[0.727, 0.830]$	$+0.055$ $[+0.018, +0.092]$
Token log-probability	0.537 $[0.469, 0.605]$	$+0.297$ $[+0.214, +0.380]$

Table 1. Perception baselines, Qwen2.5-VL-3B, $n = 300$. The upper block reduces the same five samples in different ways; the lower is a model-internal signal requiring no sampling. Differences are paired bootstraps over the same resampled items, and every one excludes zero. Entropy's margin over the simpler summaries is small but resolved, so the shape of the cluster distribution carries information that a count or a majority does not. The simple summaries are related to entropy but not redundant with it (Spearman 0.83–0.89).

4.3. What does the entropy summary add?

All five samples per item support several summaries, and a natural objection is that entropy is merely a dressed-up count of distinct answers. Table 1 tests that directly. The three sample-based alternatives are computed from the *same* generations at no additional cost, so the comparison isolates the summary rather than the sampling budget.

Two of these deserve comment. The majority margin separates a three–two split from a three–one–one split,

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Subset	n	AUROC (95% CI)
All items	300	0.835 [0.787, 0.879]
Excl. parse failures	255	0.839 [0.790, 0.886]
Excl. max-entropy items	239	0.794 [0.737, 0.848]
Excl. both	206	0.796 [0.736, 0.852]

Table 2. Perception entropy under artifact controls, Qwen2.5-VL-3B. The signal is graded rather than concentrated at the ceiling: it survives the harshest simultaneous cut with an interval excluding chance. §4.9 shows a model for which it does not.

which the majority fraction cannot, yet it remains the weakest of the three, because the information entropy exploits resides in the full distribution rather than in the top two clusters. Token log-probability is the only baseline here that is not a function of the samples, and it is also the only one that fails to beat chance; a model that is verbally or numerically confident about a page it has misread is not detectable from its own output distribution, but is detectable from asking it twice. Asking the model directly fares no better. Prompted to state a confidence from 0 to 100 alongside each transcription, it reports a median of 95 while that self-report carries no signal at all: AUROC 0.528 $[0.463, 0.593]$, against 0.807 $[0.757, 0.854]$ for entropy on the same items, a paired difference of $+0.279$ $[0.194, 0.359]$. The self-report is not degenerate, comprising 37 distinct values spanning 68 to 100. The model therefore does vary its stated confidence; it simply varies it uninformatively, which is the more damaging finding. Two independent cheap-confidence baselines, token log-probability and verbalized confidence, therefore land in the same place, and both are beaten by a resolved margin.

4.4. Does the signal survive artifact controls?

A cluster-entropy AUROC can be manufactured two ways that have nothing to do with useful uncertainty. Unparseable samples receive their own cluster label, which both inflates entropy and correlates with being scored wrong, so the metric could be measuring the parser. And items pinned at the ceiling $H = \ln K$ can carry the entire ranking if that cell happens to be uniformly incorrect, leaving a binary breakdown detector rather than a graded signal. Table 2 removes each, and both.

4.5. Does the same signal work for grading?

No. On the same 300 images, reasoning entropy predicts grading error with AUROC 0.520 $[0.458, 0.582]$, an interval containing chance. The balanced sample matters: the model answers “there is an error” for the large majority of inputs, and on an unbalanced set that bias alone would score well.

This is the honest reasoning result, and §4.6 is an account of why we did not initially believe it.

Model	Observed	Bias-only null	Collapse
Qwen2.5-VL-3B	0.854	0.901 [0.849, 0.932]	99.8%
Qwen2.5-VL-7B	0.801	0.868 [0.821, 0.903]	100.0%
LLaVA-NeXT-7B	0.775	0.831 [0.770, 0.874]	97.2%

Table 3. The apparent replication, against a null with no item-level signal at all: a model answering “error” with fixed probability per sample, so its entropy is pure sampling noise. The null *exceeds* every observed value. “Collapse” is the fraction of items on which correctness and the model’s own verdict are the same label. Even the null’s 2.5th percentile clears the 0.70 threshold we had registered in advance as confirming the hypothesis, so that threshold could never have separated signal from arithmetic.

4.6. The trap: stratifying by the label being predicted

A model with a strong answer bias makes a pooled statistic hard to read, and the natural response is to stratify by ground truth and analyse each label separately. Doing so appears to recover a large effect from the null above. Within the erroneous-solution stratum, entropy predicts grading error at 0.801 $[0.751, 0.846]$; within the correct-solution stratum it predicts in the opposite direction at 0.280 $[0.200, 0.366]$. Both strata are adequately powered, both intervals exclude chance, the sign reversal has an appealing mechanistic story, and the effect reproduces across three independently trained model families (Table 3). We reported this as our main finding.

It is an artifact, for a reason that applies to any binary-decision task evaluated this way. **Within a stratum where every item shares the same true label, “was the model correct” is not a separate fact from “what did the model answer”; the two are the same column.** On our 7B run they agree on 100% of items, and the AUROC against correctness is numerically identical to the AUROC against the model’s own verdict. Entropy is computed from precisely the votes that produce that verdict, so the measurement is near-circular. The sign reversal follows by the same identity: in the correct-solution stratum correctness is the exact *complement* of the verdict, so anything correlating with the verdict must invert. The two stratum AUROCs sum to 1.

The mechanism is arithmetic rather than behavioural. On an all-erroneous stratum the majority vote is wrong only when most samples dissent from the model’s default, and dissent is exactly what raises entropy. Nothing about the model’s competence is required for the association to appear.

Three things would have caught this earlier, and none of them is specialised. Comparing against a signal-free null rather than against chance, since 0.5 is the wrong reference

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point for a stratified statistic. Checking whether the correctness label is a relabelling of the prediction before computing anything. And treating a pre-registered threshold as necessary but not sufficient, since ours was cleared by the null.

4.7. Why transcription is not affected

Both tasks have ground truth; what differs is how many values it can take. Grading truth is binary, so stratifying by it makes the truth *constant* within each group, and correctness reduces to correct $=$ (majority vote $=$ that constant), a function of the vote, which is precisely what entropy summarises. Transcription truth is a mathematical expression drawn from an effectively unbounded set. Five samples can agree perfectly on $\frac{5}{2(x-1)}$ and still be wrong because the reference was $\frac{5}{2(x+1)}$, so agreement carries no guarantee of correctness. In the data, 38 items are unanimous and 3 of those are wrong, while 4 items at maximum entropy are right; correct and incorrect items occur at both ends of the entropy range.

It is worth being precise about which ingredient does the damage. The collapse is caused by *stratifying* rather than by the label type; grouping transcription items by their exact reference answer would degenerate in exactly the same manner. Low cardinality is what makes stratification both tempting and total: with a binary label it is the natural response to class imbalance, and it yields precisely two groups, each fully degenerate. With hundreds of distinct answers nobody would stratify, because the groups would have one member each. The trap is therefore specific to binary-decision tasks, which is where it is most likely to be walked into.

Consistent with this, §4.3 shows transcription entropy outperforming simpler summaries of the same votes; a measure that were merely counting dissent could not.

4.8. What happens on a second dataset?

The analysis in §4.5 requires a dataset containing correct solutions, and the two public alternatives we examined do not have any: in ErrorRadar [14] no item is annotated error-free, and in ScratchMath [11] the student answer never equals the reference. Neither can be balanced, and neither can exhibit the inversion.

On ScratchMath, which is the only one of the two with handwriting to grade, the measurement fails for a second and more instructive reason. Grading accuracy reads 96.0%, but every item contains an error and the model answers "error" for 90% of them, so it scores well by construction. In 24% of readings (120/500) the model explicitly states it cannot read the image, and in 82% of those (98/120) it answers "error" regardless. Excluding those readings does not clean the estimate: all four misgraded items lie among them, so the retained subset has no minority class and the AUROC is undefined rather than improved.

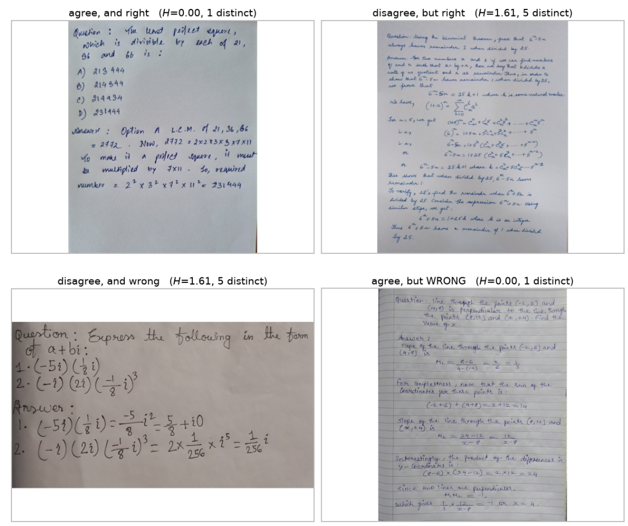

Figure 2. The four quadrants of entropy × correctness, on real pages, Qwen2.5-VL-3B. Top left is the signal working: five identical transcriptions, correct. Bottom left is it working in the other direction. Top right is a false alarm, in which the samples disagree while the majority remains correct, which is the cost of any deferral rule. Bottom right is the bound on the method: five identical transcriptions, all wrong. That quadrant is invisible to self-disagreement by construction, and it is where the audit of §6 finds the unearned passes concentrate.

We therefore report this setting as *gated*: not a negative result about uncertainty, but a statement that the pair cannot test it.

4.9. When does a high score mean nothing?

InternVL3-8B attains the highest raw perception AUROC in our study, 0.915 [0.880, 0.944], on a transcription accuracy of 21.0%, approximately half that of the model in §4.1. Independently recomputing every derived column from raw output reproduces the stored values exactly, so this is model behaviour rather than a scoring defect. Under the control of §4.4 it collapses to 0.556 [0.495, 0.614].

The mechanism is visible in the entropy distribution: 202 of 300 items sit at the ceiling, and accuracy within that group is 0.99% against 62.2% elsewhere. The score is separating total generation breakdown from everything else, which is a binary property of the output, not graded uncertainty about content. Reported without the control, it would have been the strongest number in this paper.

5. Failure Analysis

To move past anecdote we read and coded every item the frozen rule marked wrong on Qwen2.5-VL-3B, a complete census of that bucket comprising 104 of 104 items, working from the handwritten pages rather than from the extracted text. The distribution is not what the aggregate number suggests:

coded reason	n	share
scoring/extraction attributable	77	74.0%
notation misread	22	21.2%
undecidable from the page	4	3.8%
copied a different line	1	1.0%
hallucination	**0**	0%

Three things follow. First, **the model never invents content unrelated to the page**: zero hallucinations across the census, which is the failure a reviewer most expects from a VLM asked to transcribe. Second, roughly three-quarters of what the scorer calls model error is attributable to the scoring pipeline, either through the extractor selecting a wrong or partial span, or through normalization collapsing a long answer onto a single symbol. Third, and this is the constraint on the method rather than on the model, *the errors that remain are not the ones entropy can flag*: a confidently wrong item is one where all five samples agree, so self-disagreement is silent on it by construction (Fig. 2, bottom right).

Two mechanisms recur in the coded notes and both concern the final answer statement rather than the mathematics. The model verifies a solution's core computation and stops, without checking whether the final statement is complete (a variable never substituted), internally consistent (a tuple introducing z while the condition repeats x), or licensed by the problem (a unit the problem never gave). And it pattern-matches a stated result instead of recomputing it, accepting $19 \times -18 \times 23 = -7966$ where the correct value is -7866.

The coding is a single reader's, which is its main weakness. We have no second-rater agreement. We do have an intra-rater measurement: 47 items were coded twice in independent passes with different sheets, and 4 received determinate labels pointing opposite ways.

6. Limitations

The positive result rests on one dataset. It now holds on two model families (§4.2), but two further models could not test it: one sits at a transcription floor of 0.8%, and one fails the artifact control of §4.4. Two of four attempts producing a usable measurement is itself worth stating, since the result requires a model with genuine dense-OCR competence, which is not the common case among open-weight VLMs. Of the two alternative datasets, neither contains correct solutions, so neither admits a balanced sample. Moreover, by §4.6, an all-erroneous dataset constitutes a single-label stratum, on which the measurement degenerates for the same reason.

We report the retraction of §4.6 as a finding, but it was found late. It was not caught by the pre-registration, which set a threshold the null clears; nor by adequate power, which both strata had; nor by replication, which reproduced the artifact three times. Cross-model agreement is worth less than it appears when the mechanism is arithmetic rather than behavioural.

Our clustering is exact-match over an extracted answer span, not semantic. Semantic clustering was attempted and abandoned: an NLI-plus-union-find implementation merges opposed clusters through a single false-positive transitive link and degrades at larger K. The reported results avoid that path rather than depending on it being fixed.

Confidence intervals were added after the experiments were designed and run. They corrected several conclusions but cannot supply power the studies were not sized for. $K = 5$ also gives few operating points, which is why two-thirds of conformal calibration splits cannot reach a 30% risk target.

The correctness label is noisy in both directions. Every AUROC above is measured against a frozen string-matching rule, and human audit shows that rule has substantial label noise. The census of §5 finds unearned *failures*; a seeded random audit of the items the rule marked *correct* finds unearned passes as well. Correcting both directions leaves transcription accuracy an audit-sensitive range, roughly 44–61% under conservative false-pass assumptions, rather than a single figure. The two corrections act in opposing directions, and where the range settles depends on a coding judgement that the pages alone cannot resolve. The corrected AUROC is likewise a range, 0.57–0.73, depending on whether an unearned pass is treated as a model error or as undecidable. We report both as ranges because that is what the evidence supports.

This does not withdraw the perception result, and it is worth being precise about what it does. The distinct-answers relationship and the cross-family replication rest on the frozen rule's stored columns, which are unchanged. What is now in question is how well that rule's labels track human judgement. The honest statement is that **entropy predicts frozen-scorer failures; predicting human truth is harder**, and that the automatic scorer is least reliable exactly where answers are long, multi-part, set-valued, or stated as prose.

No automatic scorer we tried is safe here, including judges. We evaluated four. The frozen rule is brittle in the ways above. A span-primary variant that demotes symbolic normalization to a warning does not eliminate the unearned passes, because where both sides extract the same incorrect span, span matching agrees exactly as the normalizer did; its risk flags do, however, concentrate them. A symbolic equivalence checker is unsafe by construction on this benchmark, because the injected errors are often units, coefficients or

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audit set	n	unearned	caveat
rule said wrong	104	77	complete census; one coder
rule said correct	100	4–16	two passes disagree
flagged tier	108	39	high-risk by construction

Table 4. Human audit of the frozen scorer. **These rows must not be summed or pooled.** 78 items are shared between sets (union 234, not 312), and two of the three sets can only find errors in one direction by construction, so a pooled agreement figure would be largely tautological. The correct-side range spans two independent passes over disjoint random draws which disagree at $p = 0.00007$. That disagreement, rather than either endpoint, is what makes the corrected accuracy a range.

notation, exactly what an equivalence checker is designed to erase. Two open math judges failed differently: one rejected a valid probe outright, and the other returned the correct verdict on both probes while its own justification showed it had reconstructed a mathematically correct reference from the problem text and graded against *that* rather than against the perturbed reference it was given, misattributing the reference's answer to the model in the process. That last case is the more instructive: **a verdict-only check cannot detect it**, and on a judge that emits no reasoning it would have been recorded as a clean pass. Grading a perturbed-answer benchmark asks a judge to compare rather than to solve, and these judges solve.

7. Conclusion

On handwritten transcription, self-disagreement across repeated samples is a usable error signal: it survives every control we apply, beats the model's own confidence and simpler summaries of the same samples, and yields a deferral rule that is right 57 times in 61. On grading, the same measure carries no signal at all.

The distance between those two sentences is where the work is. Stratifying the grading task by ground truth, an ordinary response to a biased model, produces an effect of 0.80–0.85 that is adequately powered, resolves against chance, clears a pre-registered threshold, and replicates across three independently trained model families. It is still an artifact, because within a single-label stratum the correctness label is the model's own answer under another name. A signal-free biased coin scores higher.

We therefore recommend, for any evaluation of sampling-based uncertainty on a binary decision task: check whether the correctness label is a relabelling of the prediction before computing anything, and compare against a signal-free null rather than against 0.5. Neither is expensive. Both would have saved us a result we believed for three weeks.

A third caution follows from the audit in §6 and applies more broadly than to uncertainty work. On a benchmark whose reference answers are deliberately perturbed, the scorer is part of the experiment: string matching, symbolic equivalence and open math judges each fail in a different direction, and the judges fail by solving the problem rather than comparing to the reference they were handed. Where a correctness label is itself estimated, it deserves the same scrutiny as the quantity being estimated against it.

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